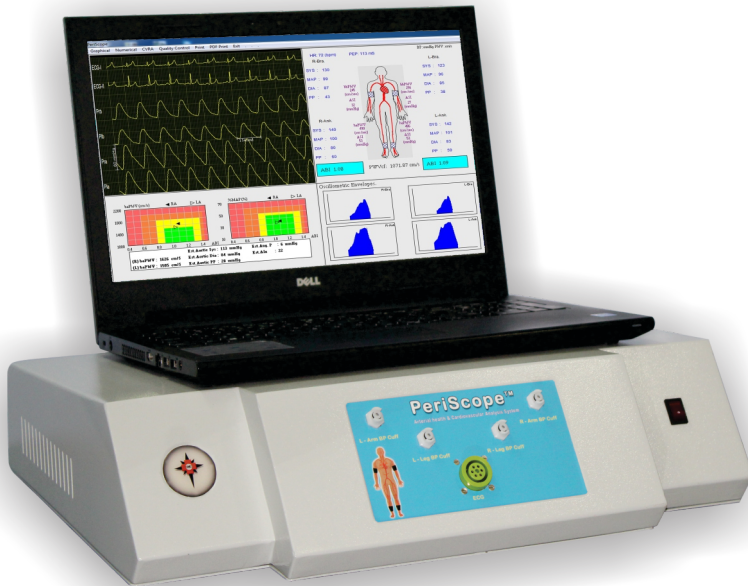




PeriScopeTM

Arterial Health & Cardiovascular Analysis System



- Central Aortic Blood pressure
- Carotid Femoral PWV (cfPWV)
- Augmentation Index
- Brachial Ankle PWV (Left & Right)
- Peripheral arterial disease

With Patented Technology#

PeriScope report provides a prognostic approach for clinicians in DM / HTN / CAD / ESRD patients.

Onset of stiffening of arteries is the beginning of vascular and later cardiovascular deterioration. This non-symptomatic /Symptomatic onset, if detected timely, can immensely help the clinician in diagnosis/ prevention / correction with a informed prognostic approach in diseases like CAD, PAD, CKD, Diabetes, Stroke & Neuropathy.

Decreased compliance of the vasculature alters arterial pressure and flow dynamics and also impacts target organ function, cardiac performance and coronary perfusion.

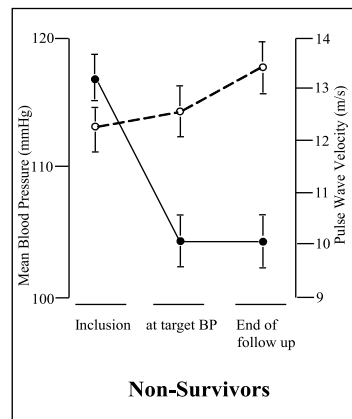
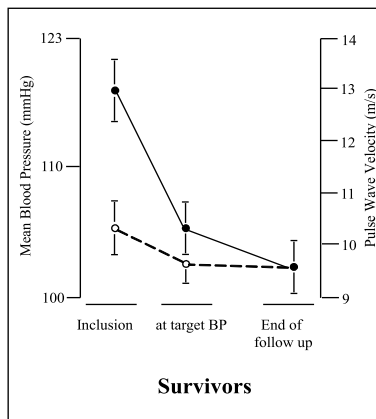
PeriScope system comprises of Windows based software and a dedicated hardware with 4 Blood pressure measuring cuffs and 4 ECG electrodes. It is very user friendly and fully automatic. Once started, the test recording completes itself by displaying results directly. The results are printed in a multicolor A4 size paper on any Windows compatible printer.

The report contains of 8-second traces of ECG, Pressure Pulse Waveforms, Nomograms and all calculated results.

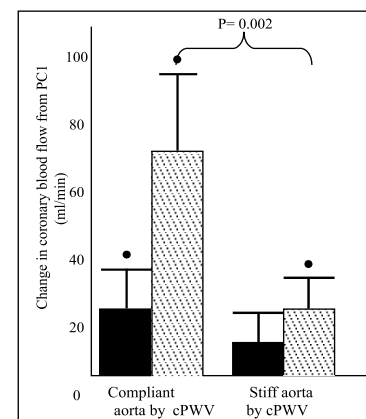
The system has a built-in database that is used to store Patient folders for further referrals/comparisons at any point of time. The results are calculated using a DSP algorithm. PeriScope computes clinical parameters: Aortic and peripheral pulse wave velocities, arterial stiffness index, pulse pressure, ankle/brachial index (ABI) and mean arterial pressure to quantify arterial stiffness.

Arterial Stiffness plays a key role in all cardiovascular diseases, be it Hypertension or Diabetes :

Since last few years, the overwhelming evidence accumulated from global research articles published in esteemed publications (from AHA, Nature, ESC, ADA, ASA, ASN etc.) in the field of cardio/vascular diseases has clearly established 'Arterial Stiffness' as the root cause of target organ damage. It has also been proven that the successful clinical outcome of a therapeutic or a surgical intervention is directly dependent on control of Arterial Stiffness of the patient.^{1,2}



Changes of MBP (solid circle) and aortic PWV (open circle) from inclusion to end of follow-up for survivors And nonsurvivors in ESRD. Values are mean +/- SEM.



■ Change in resting CBF from PCI (=2SEM) ▨ Change in hyperemic CBF from PCI (=2SEM)

CBF change in compliant & Non compliant Aorta after PCI.

There is a close association between left Ventricular Diastolic Function and Aortic Stiffness. Studies have shown that even after successful patient intervention with CABG or PTCA, if the Arterial Stiffness is not controlled, the benefits may not be satisfactory. Recurrence of acute coronary events is observed in such patients.

In essential Hypertension, Arterial Stiffness is proven as an independent predictor of fatal Stroke. Arterial Stiffness may also play a role in the pathogenesis of dementia (Vascular or Alzheimer's types).

In patients with ESRD, increased Aortic Stiffness is a strong independent predictor of all-cause and mainly Cardiovascular mortality.

Uremic patients before starting hemodialysis have significantly increased Arterial Stiffness as well as those on maintenance hemodialysis.

Macrovascular disease is the major cause of morbidity and mortality in patients with type 2 diabetes. Arterial stiffness plays a critical role in the development of Atherosclerosis, a process which occurs prematurely and at an accelerated rate in patients with diabetes.

Arterial stiffness is a powerful independent predictor of later mortality across the entire spectrum of glucose tolerance, with or without over type 2 diabetes. It reflects a final common pathway on which BP and other risk factors operate.

Arterial Stiffness measurement with PeriScope™

Prevention, detection, therapeutic reversal and surgical intervention follow-up of Arterial Stiffness is important in all stages of Cardiovascular diseases to the extent that Arterial Stiffness measurement is a recommended routine clinical test^{3,4}.

Drugs like ACE-inhibitors, Calcium-channel antagonists, Statins, AGE-breakers, ARBs, Nebivolol etc. are known to treat the increasing Arterial stiffness. But to find out their efficacy for a particular patient and titrate the dosage, a periodic quantitative measurement of Arterial Stiffness is required.

Arterial Stiffness measurement with PeriScope™ is cost-effective, easy and quick. The test report with Cardiovascular Risk Analysis (CVRA) gives all the necessary details like Parameter Normal Values, Observed Values and Interpretive Significance.

PeriScope™ Parameters and Arterial stiffness analysis : A Typical Case with Optimal BP

In the following patient , it may be noted that his BP and HR are optimal. PeriScope test conducted on this patient reveals that his left (ankle) baPWV is very high than the predicted normal. This may be seen as a manifestation of arterial stiffness into beginning of PAD. His C-F PWV is High, i.e. his aortic stiffness has increased. Despite the optimal BP, this arterial stiffness subsequently will lead to CAD if left untreated and compromise the coronary blood flow and perfusion pressure. In short, this patient is already undergoing a functional change, if treated properly and in a timely manner this functional change can be arrested and/or reversed.

Parameter	Unit	Normal Value	Observed Value	Significance*
Heart Rate	BPM	60 to 100	70.9	HR is Normal.
Blood Pressure	mmHg	120 / 80	113 / 68	Optimal Blood Pressure
Pulse Pressure	mmHg	Less than 50	45	Normal Pulse Pressure
Right baPWV	cm/s	Less than 1263.86	1377.9	Mild CA Stiffness
Left baPWV	cm/s	Less than 1263.86	1709.8	Very High CA Stiffness
C-F PWV	cm/s	Less than 787.30	1053.2	High Aortic Stiffness
R Bra ASI	mmHg	20 to 70	22.2	Normal rbASI
L Bra ASI	mmHg	20 to 70	19.4	Normal lbASI
R Ank ASI	mmHg	20 to 70	37.4	Normal raASI
L Ank ASI	mmHg	20 to 70	33.4	Normal laASI
R - ABI	--	0.9 to 1.35	1.23	ABI Normal:Peripheral ASO may be absent
L - ABI	--	0.9 to 1.35	1.27	ABI Normal:Peripheral ASO may be absent
Ao Sys	mmHg	Less than 116	100	Normal Aortic Sys Pressure
Ao PP	mmHg	Less than 34	32	Normal Aortic Pulse Pressure
Ao Dia	mmHg	Greater than 65	68	Normal Aortic Dia Pressure
Ao AugP	mmHg	Less than 6	7	Borderline increase in Ao Aug P
Alx (Alx@HR75)	%	Less than 12(13)	23(24)	Moderate increase in Ao Alx

* This may be Clinically Correlated

Abbr.: CA - Central Arterial, PA - Peripheral Arterial, PWV - Pulse wave velocity, baPWV - Brachial Ankle PWV, C-F PWV - Carotid Femoral PWV
ASI - Arterial Stiffness Index, ASO - Atherosclerosis obliterans, Ao - Aortic, Aug - Augmentation, Alx - Augmentation Index

Major Parameters :

Pulse Wave Velocities- Carotid Femoral Pulse wave velocity is the GOLD STANDARD measurement for central Arterial Stiffness⁶. Segmental pulse wave velocities to find Peripheral Arterial Stiffness. Increase in PWV is definitive indicator of increase in arterial stiffness.

Arterial Stiffness Index- ASI is an indicator of Peripheral Arterial Stiffness and a measure of pulsatility for lower limb arteries.

Ankle Brachial Index- ABI is a standard measurement for lower limb atherosclerosis.

Mean Arterial Pressure- MAP is a standard measurement for Arterial compliance,.

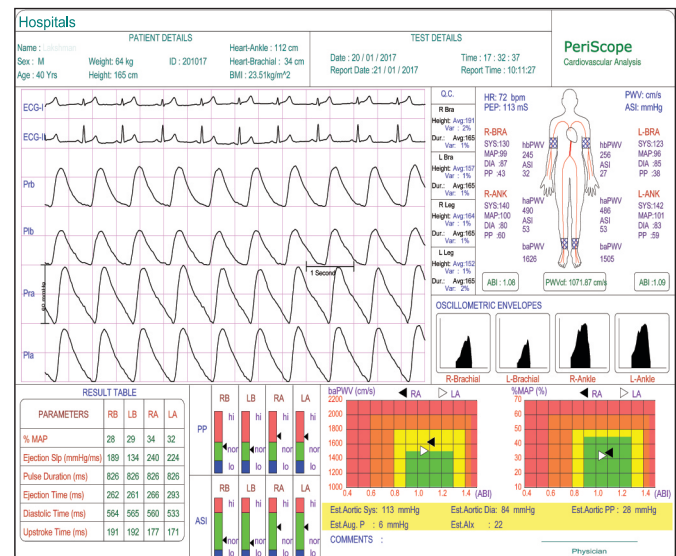
Pulse pressure- Indicator of aging of arteries. Stiffer arteries have wider pulse pressure.

Percentage MAP- Lower % is an indicator of decrease in compliance of Arteries.

All these parameters are established independent markers to analyze the Arterial Stiffness, Atherosclerosis and its organ oriented effects.

PeriScope™ is clinically validated as per International guidelines with relevant publications in International and National journals.⁶

The Normal Values are tested and correlated in Indian population.⁷



PeriScope™ Overview & Test Procedure

The patient in supine position is attached with 4 limb BP cuffs and a 4 lead ECG cable to the PeriScope hardware. After entering all the patient details like age, sex, weight, height the test is started. This test is completely automatic. In the first run, all the 4 cuffs inflate and deflate recording the Systolic/Diastolic Blood Pressures and Mean Arterial Pressure. After a brief pause, the second run starts during which the blood pressure oscillations are recorded simultaneously for all the 4 limbs along with the ECG waveforms. The total time required is about 10 minutes for the complete procedure including attachment of sensors.

Using these waveforms, various parameters like pulse wave velocities, ABI, Nomograms are calculated. All the results are displayed simultaneously on the screen. The software also has a unique quality control to validate the conducted test. It helps to reconduct the test properly if the consistency in acquired waveform is not enough to calculate accurate results.

The on-screen analysis consists of all the acquired waveforms, Pulse Wave Velocities, BP values, ABI values, Graphical interpretive representation of the results, tabular representation of parameters, observed and normal values and their significance. This analysis gives a comprehensive diagnostic picture of the patient's cardiovascular status at a single glance.

These test results are also available for review through a database for future reference and comparative assessment.

References

1. Aortic Stiffness Affects Coronary Blood Flow to Percutaneous Coronary Intervention: Michael C H Leung, et al: Am j Physiol Heart Circ Physiol 290: H624-H630, 2006
2. Impact of Aortic Stiffness Attenuation on survival of Patients in End-stage renal Failure: Guerin et al: Circulation (AHA Journal) 2001;103:987-992.
3. 2007 guidelines for management of hypertension : ESC and ESH guidelines : European Heart journal (2007) 28,1462-1536.
4. Expert consensus document on arterial stiffness: methodological issues and clinical applications Stephane Laurent et al : European heart Journal (EHJ-ESC Journal) doi.10.1093/eurheartj/ehl254
5. As above.
6. Validity and reproducibility of arterial pulse wave velocity measurement using new device with oscillometric technique: A pilot study: MUR Naidu et al : Biomedical Engineering Online 2005, 4:49

Non-Invasive evaluation of arterial stiffness in patients with increased risk of cardiovascular morbidity: A cross-sectional study : Sridhar et al: Indian J pharmacol 2007 Volume: 39 Issue: 6 Page: 294-298.

Specifications[§]

Blood Pressure

Method: Oscillometric
Pressure Detection: Semiconductor Sensor
Pressure Display Range: 40-240 mmHg

ECG

Continuous ECG Waveform
Sensing Method: Direct contact 4 electrodes
Heart Rate Range: 30-240bpm

Test Method

Fully automatic Oscillometric Recordings on all limbs simultaneously

Database

MSAccess Easy recall of any previous test

Waveforms

ECG Lead I
ECG Lead II
Right Arm BP
Left Arm BP
Right Ankle BP
Left Ankle BP

Interpretation

Cardio Vascular Interpretation
Clinical Recommendations
Therapeutic Suggestions

Minimum PC Requirements

OS : Windows* 7 / 8 / 10
CPU : Celeron* 1.8 GHZ / i3 / i5 or better processor
HDD : 500GB
RAM : 2GB
Printer : Any Windows* Compatible Inkjet / Laser Jet Printer

Power Requirement and other details

Voltage 230 VAC $\pm$ 10%, Frequency 50 Hz,
Power consumption 20VA max
Instrument Dimensions : 500 mm X 400 mm X 100 mm
Approximate Instrument Weight : 9.60 Kg

ENVIRONMENT REQUIREMENTS

Operating and storage Temp. 15°C to 45°C
Ambient Relative Humidity 15% to 90%, non- condensing
Operating and storage Pressure/Altitude 523 mm Hg to 760 mm Hg

§ Specifications are subject to change without prior notice

*All regd. Trademarks are acknowledged to their respective owners.

#PeriScope™ uses technology patented by Genesis Medical Systems Pvt. Ltd. as "APPARATUS AND METHODS FOR THE NON-INVASIVE MEASUREMENT OF AORTIC PRESSURE PARAMETERS AND CARDIOVASCULAR SYSTEM PARAMETERS" under Patent No.: 341138

Since R&D is a continuous process, features & specifications are subject to change without notice.

Certifications | CE | EN ISO 13485:2016

Manufactured by :

Genesis Medical Systems Pvt. Ltd.

#5-9-59, 5th Floor, Moghul's Court Building,
Basheerbagh, Hyderabad - 500 001. Telangana (INDIA)
Ph.: +91-40-23236155 / 23236165
www.genesismedicals.com
e-mail: sales@genesismedicals.com

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